Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Simplification of algebraic expressions: order of operations, brackets and word problems. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Second Term
Week	9
Topic	Simplification of Algebraic Terms
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems
Theme	—
Curriculum objectives	• carry out order of operation in algebra • insert/remove bracket and simplify expression • solve word problems involving algebraic expression
Previous knowledge	• Addition and subtraction of algebraic terms • Multiplication and division of algebraic terms
Lesson sequence	• carry out order of operation in algebra — the teacher solves some questions in order of operation on the board; the student listen, contribute, and where necessary ask questions where they do not understand; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$ • insert/remove bracket and simplify expression — the teacher solves some questions on insertion, removal of bracket on the board; the student listen, ask questions on their areas of difficulty; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, marks their notes and makes corrections. Set for the students: (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$ • solve word problems involving algebraic expression — the teacher solves some questions on word problems involving algebraic expression on the board; the student listen, ask questions where they do not understand; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.
Instructional materials	• Chart showing algebraic expression.
Evaluation	• Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$ • Simplify $7 \times 3a - (3a + 5a) \times 2$

	• Simplify $12-(7-2f)-(3-f)$ • Simplify $(8a-5)+(14-3a)$ • A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Lesson objectives

1.	carry out order of operation in algebra
2.	insert/remove bracket and simplify expression
3.	solve word problems involving algebraic expression

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Simplification of algebraic expressions: order of operations, brackets and word problems

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: Carrying Out Order of Operations in Algebra

When you simplify an algebraic expression, the terms must be dealt with in a set order. You already know how to add, subtract, multiply and divide algebraic terms; the order of operations simply tells you *which* operation to do *first* when several appear in one expression.

Work through the operations in this order:

1. **Brackets** — work out anything inside brackets first.
2. **Division and Multiplication** — do these next, working from left to right.
3. **Addition and Subtraction** — do these last, working from left to right.

The important point for algebra is that **multiplication and division are always done before addition and subtraction**. A very common mistake is to add terms first because they look alike — but the order of operations must be respected first.

For example, in 4n + 3n × 10, you must NOT add 4n + 3n first, because the multiplication 3n × 10 has to come before the addition. Deal with the multiplication, then add what is left.

Example 1 — Simplify 4n + 3n × 10

Multiplication comes before addition:
3n × 10 = 30n

Now add the like terms:
4n + 30n = 34n

So 4n + 3n × 10 = 34n.

Example 2 — Simplify 32 + 8x ÷ 2

Division comes before addition:
8x ÷ 2 = 4x

Now add:
32 + 4x

32 and 4x are not like terms, so they cannot be combined. The answer stays as 32 + 4x.

Example 3 — Simplify 3 × 5x + 4x ÷ 2

Do the multiplication and the division first, working from left to right:
3 × 5x = 15x and 4x ÷ 2 = 2x

Now add the like terms:
15x + 2x = 17x

So 3 × 5x + 4x ÷ 2 = 17x.

Example 4 — Simplify 36y ÷ 6 + y + 2 × 4y

Do the division and the multiplication first, from left to right:
36y ÷ 6 = 6y and 2 × 4y = 8y

Now add all the like terms:
6y + y + 8y = 15y

So 36y ÷ 6 + y + 2 × 4y = 15y.

Practice

1. Simplify 4n + 3n × 10.
2. Simplify 32 + 8x ÷ 2.
3. Simplify 3 × 5x + 4x ÷ 2.

4. Simplify 36y ÷ 6 + y + 2 × 4y.
5. Simplify 7 × 3a − (3a + 5a) × 2.

Section 2: Removing Brackets to Simplify

Brackets in algebra group terms together so we can treat them as one piece. To simplify an expression that contains brackets, you first remove the brackets and then collect like terms. Everything depends on the sign written directly in front of the bracket.

• If a plus sign (+) is in front of the bracket, remove the bracket and leave every sign inside exactly as it is.
• If a minus sign (−) is in front of the bracket, remove the bracket and change every sign inside: + becomes −, and − becomes +.

Only the sign in front of the bracket matters — a number written right against a bracket (like $3(2a+1)$) means multiplication, which is a separate idea. Here, every bracket is added or subtracted as a whole.

You can also work the other way and insert a bracket. If the bracket you put in has a plus sign before it, keep the signs of the grouped terms as they are. If it has a minus sign before it, change the signs inside. For example, $3x-y+4$ can be written as $3x-(y-4)$, because removing that bracket gives $3x-y+4$ again.

Simplify $(12a-7)+(4-5a)$.

There is a plus sign in front of each bracket, so remove both brackets without changing any sign inside:

$$12a-7+4-5a$$

Now collect like terms. The terms with a are $12a$ and $-5a$; the constant terms are -7 and $+4$.

$$12a-5a=7a \quad \text{and} \quad -7+4=-3$$

So the answer is:

$$7a-3$$

Now a case with a minus sign. Simplify $(8x-2y)-(9x-4y)$.

The first bracket has a plus (implied) in front, so it is removed as it is. The second bracket has a minus in front, so every sign inside is changed: $9x$ becomes $-9x$ and $-4y$ becomes $+4y$.

$$8x-2y-9x+4y$$

Collect like terms:

$$8x-9x=-x \quad \text{and} \quad -2y+4y=2y$$

So the answer is:

$$-x+2y$$

Simplify $12-(7-2f)-(3-f)$.

The first 12 has no bracket. The bracket $(7-2f)$ has a minus in front, so change its signs: 7 becomes -7 and $-2f$ becomes $+2f$. The bracket $(3-f)$ also has a minus in front, so 3 becomes -3 and $-f$ becomes $+f$.

$$12-7+2f-3+f$$

Collect like terms. For the constants: $12-7-3=2$. For the f-terms: $2f+f=3f$.

$$3f+2$$

You can also insert a bracket. Write $3x-y+4$ with the last two terms inside a bracket that has a minus in front. To keep the signs right when the bracket is removed, put $y-4$ inside:

$$3x-(y-4)$$

Check by removing the bracket: the minus changes y to $-y$ and -4 to $+4$, giving $3x-y+4$, which matches the original.

Practice

1. Simplify $(8a-5)+(14-3a)$.
2. Simplify $(5m+8)-(2m+3)$.
3. Simplify $14-(3-4k)-(2+k)$.

Section 3: Turning Word Problems into Algebraic Expressions

Sometimes a problem is given to you in words rather than as a ready-made expression. Your job is to read the words carefully, work out which quantity each letter stands for, and then build an algebraic expression from what you are told.

The key is to decide two things:
1. What does each letter represent?
2. How do the words describe a relationship between the quantities?

Words like "more than" tell you to add, "less than" tells you to subtract, and "altogether" or "total" tells you to add things together. Work step by step, and write down what each letter stands for before you try to build the expression.

Worked example

A pen costs h naira, and a crayon costs ₦20 less than a pen.

(a) Find the cost of a crayon.

First, decide what the letter stands for: h is the cost of a pen in naira.

A crayon costs ₦20 *less* than a pen, so we subtract 20 from the cost of the pen:

$$\text{cost of a crayon} = h - 20$$

(b) Find the cost of a crayon and a pen.

"and" here means we add the two costs together. The pen costs h naira and the crayon costs $h - 20$ naira:

$$\text{cost of a crayon and a pen} = h + (h - 20)$$

Now simplify by removing the bracket and collecting like terms:

$$h + h - 20 = 2h - 20$$

So a crayon costs $(h - 20)$ naira, and a crayon together with a pen costs $(2h - 20)$ naira.

Worked example

A pen and a pencil altogether cost x naira. If a pen costs y naira, find the cost of 3 pencils.

Step 1 — find the cost of one pencil.

"Altogether" means the pen and pencil added together make x naira. The pen costs y naira, so:

$$y + \text{cost of one pencil} = x$$

To get the cost of one pencil, subtract y from both sides:

$$\text{cost of one pencil} = x - y$$

Step 2 — find the cost of 3 pencils.

Three pencils cost three times one pencil:

$$3 \times (x - y)$$

So 3 pencils cost $3(x - y)$ naira, which we could also write as $3x - 3y$.

Practice

1. Envelopes cost d naira per pack. Four notepads cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelopes and one notepad.
2. A book costs b naira and a ruler costs ₦10 less than the book. What is the total cost of 2 books and 3 rulers?
3. A pen and a pencil altogether cost x naira. If a pen costs y naira, find the cost of 3 pencils. (Simplify your answer.)

Check your work

Attempt every question before you read this page.

Carrying Out Order of Operations in Algebra

1. 34n

   Multiplication comes before addition: 3n × 10 = 30n, then 4n + 30n = 34n.
2. 32 + 4x

Division comes before addition: 8x ÷ 2 = 4x, so the expression is 32 + 4x. 32 and 4x are not like terms, so they cannot be combined.

3. 17x

   Do the multiplication and division first, left to right: 3 × 5x = 15x and 4x ÷ 2 = 2x. Then add the like terms: 15x + 2x = 17x.

4. 15y

   Do the division and multiplication first, left to right: 36y ÷ 6 = 6y and 2 × 4y = 8y. Then add all the like terms: 6y + y + 8y = 15y.

5. 5a

   Brackets come first: 3a + 5a = 8a. Now the expression is 7 × 3a − 8a × 2. Do the multiplications next, left to right: 7 × 3a = 21a and 8a × 2 = 16a. Then subtract: 21a − 16a = 5a.

Removing Brackets to Simplify

1. $5a + 9$

   Both brackets have a plus in front, so remove them without changing signs: $8a - 5 + 14 - 3a$. Collect like terms: $8a - 3a = 5a$ and $-5 + 14 = 9$. Answer: $5a + 9$.

2. $3m + 5$

   The first bracket has a plus in front, so keep its signs: $5m + 8$. The second bracket has a minus in front, so change its signs: $2m$ becomes $-2m$ and 3 becomes -3. This gives $5m + 8 - 2m - 3$. Collect like terms: $5m - 2m = 3m$ and $8 - 3 = 5$. Answer: $3m + 5$.

3. $3k + 9$

   The first bracket $(3 - 4k)$ has a minus in front, so change its signs: 3 becomes -3 and $-4k$ becomes $+4k$. The second bracket $(2 + k)$ has a minus in front, so 2 becomes -2 and k becomes $-k$. This gives $14 - 3 + 4k - 2 - k$. Collect like terms: $14 - 3 - 2 = 9$ and $4k - k = 3k$. Answer: $3k + 9$.

Turning Word Problems into Algebraic Expressions

1. A pack of envelopes costs d naira. One notepad costs $\frac{d+20}{4}$ naira. Total cost of a pack of envelopes and one notepad is $d + \frac{d+20}{4}$ naira.

   Four notepads cost 20 naira more than a pack of envelopes, which is $d + 20$ naira. So one notepad costs $\frac{d+20}{4}$ naira. Adding the pack of envelopes gives $d + \frac{d+20}{4}$ naira.

2. $2b + 3(b - 10) = 2b + 3b - 30 = 5b - 30$ naira.

   A ruler costs $b - 10$ naira. Two books cost $2b$, and three rulers cost $3(b - 10) = 3b - 30$. Total is $2b + 3b - 30 = 5b - 30$ naira.

3. One pencil costs $x - y$ naira, so 3 pencils cost $3(x - y) = 3x - 3y$ naira.

   Altogether the pen and pencil make x naira, and the pen costs y naira, so one pencil costs $x - y$ naira. Three pencils cost $3(x - y)$, which simplifies to $3x - 3y$ naira.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-687c30df	**Date**	
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____